Prodigy Anesthesia Quick Review

Addisons Disease

What are the symptoms of Addison's disease?	Weakness, anorexia, nausea, vomiting, hyperpigmentation, chronic hypotension, hypovolemia, hyponatremia, and hyperkalemia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 398-400.*
What are the two types of adrenal insufficiency?	There are two classifications of adrenal insufficiency: primary and secondary. In primary adrenal insufficiency the adrenal glands cannot produce enough hormones. Secondary adrenal insufficiency is due to suppression or disease of the hypothalamic/pituitary axis. Addison's disease is the idiopathic autoimmune destruction of the adrenal gland and is a form of primary adrenal insufficiency. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 398-400.*
How do the effects of primary adrenal insufficiency differ from secondary insufficiency?	Primary adrenal insufficiency results in the inadequate release of glucocorticoid, mineralocorticoid, and androgen hormones. Secondary adrenal insufficiency results in the inadequate release of glucocorticoid only. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 398-400.*
What are the most common causes of secondary adrenal insufficiency?	Iatrogenic causes are the most common and include administration of synthetic glucocorticoids and pituitary surgery or irradiation. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 398-400.*
What patients are at risk for adrenal insufficiency?	Any patient on chronic exogenous steroid therapy is at risk for adrenal insufficiency during periods of stress such as surgery. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 398-400.*
What plasma cortisol level is diagnostic of adrenal insufficiency?	Plasma levels less than 20 mcg/dL indicate adrenal insufficiency *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 398.*
How does general anesthesia affect activation of the HPA axis? How does regional anesthesia affect it?	Deep general anesthesia and regional anesthesia both postpone the increase in plasma cortisol secretion until the postoperative period. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 398-400.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.

Prodigy Anesthesia Quick Review

Addisons Disease

What are the surgical factors that influence the activation of the hypothalamic-pituitary-adrenal axis? How does surgery affect it?	The magnitude and duration of surgery determine the degree to which the HPA axis is stimulated. Increases in adrenocorticotropic hormone begin immediately after incision, remain elevated during surgery, and peak at the time neuromuscular blockade is reversed. Major surgery can elevate plasma cortisol secretion from 15 to 25 mg/day to 75 to 150 mg/day. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 398-400.*
How long after abrupt withdrawal of steroids would symptoms begin to appear in a patient on a long-term exogenous steroid regimen?	Symptoms would begin to appear within 24 to 36 hours. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 398-400.*
How long after discontinuation of long-term steroid use will it take for adrenal function to return to normal?	It can take 6 to 12 months for adrenal function to return to baseline after chronic exogenous corticosteroid use. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 398-400.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.